

Matteo Cherubini

Front-end developer • Applied AI and knowledge systems

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ABOUT ME

Front-end developer (Angular) with three years' experience building the digital backbone of Italian education and cultural heritage: HUB Scuola, the main learning platform for Italian schools, and platforms for the National Central Library of Rome and CNR research infrastructures linked to the European Open Science Cloud — with a consistent focus on web accessibility.

Since May 2026 an independent ICT consultant in applied AI and knowledge management, developing and piloting systems that combine a security model, local LLM inference and human review. I work equally well inside structured teams and in full autonomy, including projects for regional public bodies run from requirements to delivery. English at C1 level (CEFR, self-assessed).

My method is algorithmic: break the problem down, optimise the flow, and document every decision so that the work can be reviewed, handed over and maintained.

WORK EXPERIENCE

1 May 2026 – Current Prato, Italy	ICT consultant <i>Self-employed (VAT-registered freelance activity)</i> Independent technology consultant in applied AI and knowledge management, building local-first systems in which automation is paired with a security model and human review. • Developing and piloting AI-assisted workflow automation with an accountancy firm (pilot in progress), to reduce repetitive document handling and free professional time for analysis and judgment. • OCR pipelines: turning scanned and digital documents into structured, searchable text that feeds the downstream automation. • Dynamic wikis: documentation that an LLM agent keeps up to date as new sources arrive, so that knowledge is compiled once and maintained, rather than re-derived at every query as in the standard RAG approach. • Knowledge Genome Orchestrator: the in-house system behind this work: a distributed, encrypted, multi-domain knowledge base (no vector database, no embedding pipeline, no retrieval server), designed to extend to environmental, agronomic and financial domains. Two-phase agent architecture with an explicit trust boundary: the agent performs semantic work only, and a deterministic, tested post-processor validates its output before anything touches the filesystem. • Security and quality model: per-domain AES-256 encryption (git-crypt) with keys injected at runtime from a password manager and never written to disk; human review gate through pull requests; deterministic test suite with no dependency on model, GPU or network; multi-domain layout on Git submodules, orchestrated via webhooks and local LLM inference.
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- **Self-hosted infrastructure:** Docker Compose stacks (chosen over Kubernetes for operational simplicity and fast recovery) with reverse proxy and SSL management, private Git hosting (Forgejo), workflow automation (n8n with PostgreSQL and Redis), secure external access via Cloudflare Tunnel, and local LLM inference (Ollama) on NVIDIA GPU; reproducible, independently deployable stacks.
- Tools: TypeScript on Bun, Docker Compose, Proxmox, Forgejo, n8n, Ollama, git-crypt, Vaultwarden, Obsidian; systems co-designed with state-of-the-art LLMs.

24 Jan 2023 – 1 May 2026

Florence and Pisa, Italy

Front-end developer

Gruppo Meta / M.E.T.A. S.r.l. (ETT Group), later Deda Next S.r.l. (Dedagroup)

Front-end developer (Angular) in a software house specialised in the digitalisation of educational publishing, cultural heritage and research. Permanent contract; work split between the Florence and Pisa offices.

- Angular development for Mondadori Education's HUB Scuola, the main digital learning platform for Italian schools: rollout of the new design system and independent development of new features, in particular interactive quizzes and exercises embedded in e-books.
- Translation of design mock-ups into Angular components; design and implementation of entire application flows of varying complexity.
- Web accessibility (a11y) in line with EU Directive 2016/2102; refactoring, debugging and unit testing.
- Large-scale modernisation of legacy viewers for complex editorial formats, carried out autonomously in close cooperation with the design and UX teams.
- Front-end contribution, within project teams, to platforms for national cultural and research institutions: digitalisation of the collections of the National Central Library of Rome (metaCMS suite, interoperable with Europeana), and CNR research infrastructures in the digital humanities under the H2IOSC project, funded by NextGenerationEU (Italian NRRP) and part of the Italian contribution to the European Open Science Cloud (EOSC).
- Projects for regional public bodies managed end-to-end, from requirements gathering to delivery, with direct contact with the client.
- Development and integration of Node.js APIs together with the back-end team; data visualisation (charts and tables).
- Tools: Angular, TypeScript, Node.js, Docker, Postman, Figma, Git (structured repositories with submodules).

EDUCATION AND TRAINING

1 Oct 2016 – 7 Jun 2022

Florence, Italy **EQF 6**

Laurea in Informatica (Bachelor's Degree in Computer Science) — Class L-31, Computer Science and Technologies

University of Florence — School of Mathematical, Physical and Natural Sciences

Three-year degree in Computer Science (180 ECTS).

- Programming and software engineering: programming, programming methodologies, concurrent programming, algorithms and data structures, interpreters and compilers, theoretical computer science.
- Systems and networks: operating systems, computer architecture, computer networks.
- Data and mathematics: databases and information systems, probability and statistics, numerical analysis, calculus I and II, linear algebra, discrete mathematics and logic, general physics.
- Security: computer forensics — digital investigation and the legal framework of IT evidence.
- Design and business: multimedia design and production, business skills.

- Curricular internship (12 ECTS).
- Final thesis: “Progressive Web Applications and Web Operating Systems” (supervisor: Prof. Rosario Pugliese; co-supervisor: Prof. Francesco Tiezzi). An analysis of Progressive Web Apps as a standards-based, cross-platform alternative to native applications, including a fully working PWA developed with Node.js, Ionic, Vue.js and Firebase.

22 Aug 2022 – 12 Oct 2022

Indianapolis, United States

Program for Intensive English (PIE) — Level 5 of 7 completed

Indiana University–Purdue University Indianapolis (IUPUI) — IU School of Liberal Arts

Intensive academic English programme (Fall 1 session, minimum 140 hours of classroom instruction) at an American university.

- Completed Level 5 of 7 in Reading–Writing and Listening–Speaking, grade average 88/100 (PIE Letter of Achievement, 13 October 2022).
- Academic writing, oral presentation and discussion skills.
- Full immersion in an international student environment.

28 Oct 2025 – Current
Prato, Italy

Sommelier training course — Level 1 completed, Level 2 in progress

Associazione Italiana Sommelier (AIS) — Tuscany

Professional sommelier training: viticulture and oenology, tasting technique and sensory analysis, Italian and international wine regions, food and wine pairing, wine service.

PROJECTS

2026 – Current

NETKIT — network field audit toolkit

Personal project

A terminal application for professional network auditing, designed for consultants and freelancers, in which the legal layer is a first-class component rather than an afterthought.

- Digital consent flow producing a non-repudiable SHA-256 hash of agreement, timestamp and scope.
- Fail-closed “scope guard” validating every IP and subnet in real time, with time windows and exclusion lists.
- Alignment with NIS2 and GDPR; local-first AI reporting so audit data never leaves the machine.
- Built on Bun and OpenTUI with strictly typed TypeScript (branded types, result pattern) and SQLite.

SKILLS

Front-end and web

TypeScript · Angular · CSS · JavaScript · implement front-end website design · design user interface · World Wide Web Consortium standards · use markup languages

AI and emerging technologies

principles of artificial intelligence · emergent technologies · use automatic programming

Programming and software engineering

use software design patterns · develop software prototype · object-oriented modelling · use object-oriented programming · use software libraries · debug software

Security and compliance

ICT security legislation · NIS2 and GDPR alignment · digital forensics fundamentals

Infrastructure and tools

tools for software configuration management · Git · Docker · Forgejo · n8n · Ollama · Proxmox

Client and consulting

identify customer requirements · propose ICT solutions to business problems · collect customer feedback on applications · use online tools to collaborate · adapt to changes in technological development plans · keep up with the latest information systems solutions

LANGUAGE SKILLS

Mother tongue

Italian

English

C1 (CEFR self-assessment). Program for Intensive English, IUPUI, Indianapolis, August–October 2022: Level 5 of 7 completed in Reading–Writing and Listening–Speaking.

LISTENING
C1

READING
C1

WRITING
C1

SPOKEN PRODUCTION
C1

SPOKEN
INTERACTION
C1

NETWORKS AND MEMBERSHIPS

1 Oct 2025 – Current

Associazione Italiana Sommelier (AIS) — Tuscany

DRIVING LICENCE

11 Jul 2016 – 27 Mar 2034

Category B

9 Sep 2023 – 27 Mar 2034

Category A

HOBBIES AND INTERESTS

Agriculture, food and beverage culture

Sustainable, non-intensive agriculture and quality food supply chains (Slow Food), a topic I also follow from within the family agricultural business. Viticultural biodiversity through G.R.A.S.P.O., the ampelographic research association recovering ancient and rare grape varieties. Wine culture (AIS member, sommelier in training). Specialty coffee.

Sustainable development, finance and society

Ethical finance, microcredit and ESG investing. Geopolitics, accessibility and digital rights.

ADDITIONAL INFORMATION

Family agricultural business

Managing director and legal representative of Casal de' Cherubini S.r.l., a family-owned agricultural company managing farmland and rental property in Umbria. Signatory role exercised in agreement with the other family members, with no day-to-day operational involvement.

Curriculum Vitae — Matteo Cherubini · Structured after the Europass format

Set in Atkinson Hyperlegible · Last updated: September 2026